



ML-Powered Medical Diagnosis Assistant

Rachith Bharadwaj T N, Dhanush N, Kishan Kumar Y, Shreedhar M Kadakol

Agenda

01

The Challenge

Identifying critical gaps in current diagnostic workflows.

02

Our Solution

An interpretable ML-driven assistant.

03

Technical Architecture

Deep dive into core components and technologies.

04

Data & Validation

Ensuring accuracy and reliability.

05

Impact & Next Steps

Vision for deployment and future enhancements.

Key ML-Based Solution Components



Symptom-to-Disease Classification

Utilizing interpretable models like Random Forest, XGBoost, or Logistic Regression for clear diagnostic pathways.



EHR Pattern Matching

Leveraging Apache Spark MLlib for efficient processing and pattern identification within Electronic Health Records.



Offline Speech-to-Text

Integrating Vosk or CMU Sphinx for robust voice interaction without internet dependency.



Statistical Validation

Ensuring model reliability through rigorous hypothesis testing and confidence intervals.

Dataset Strategy for Model Training & Validation

Our solution's accuracy is underpinned by a diverse and robust dataset strategy.



Disease & Symptoms (Kaggle)

Primary dataset for supervised learning, enabling initial model training and classification.



MIMIC-IV / eICU

Critical for Spark MLlib-based validation and pattern matching against real-world EHR data.

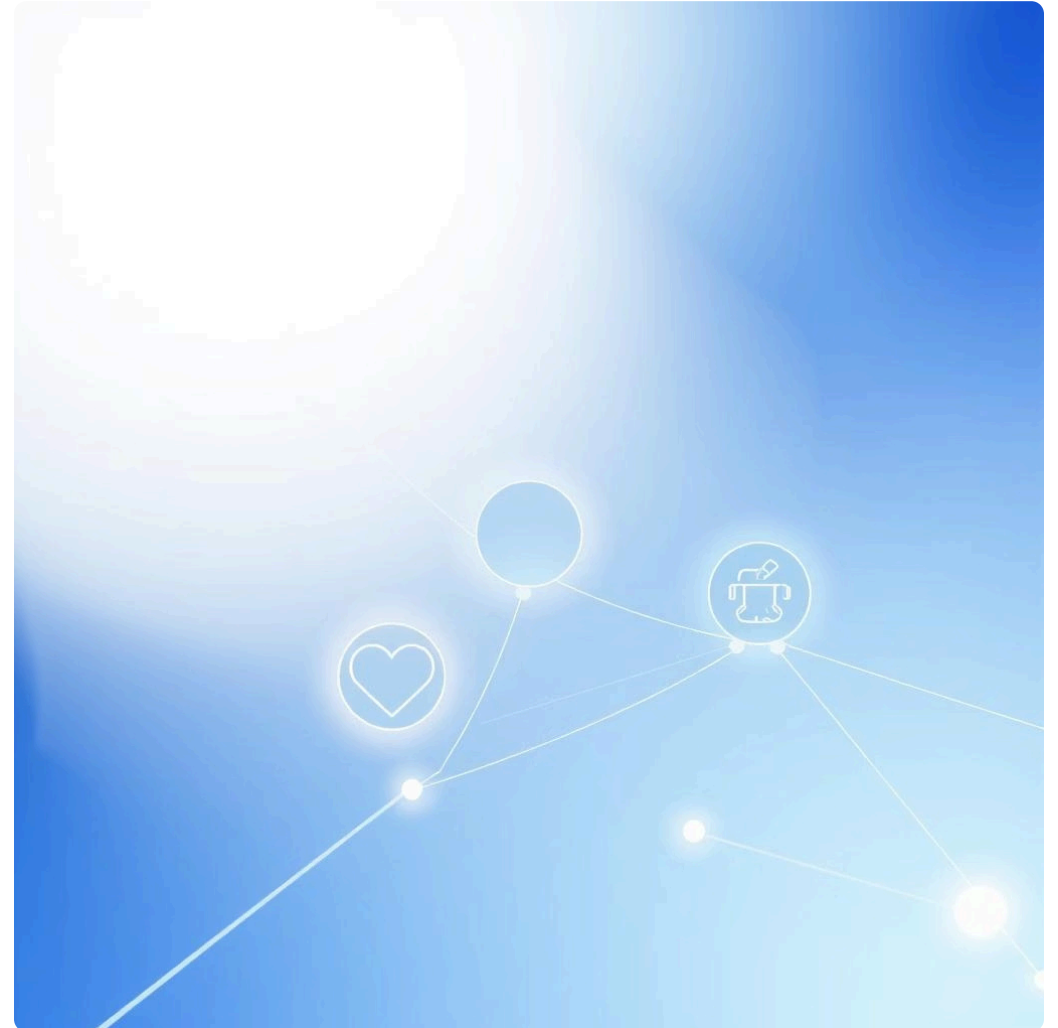


Open Disease Symptoms CSV

Used for rapid prototyping and developing the offline logic, ensuring broad applicability.

Expected Output: Empowering Clinicians & Patients

- **Increased Diagnostic Accuracy:** Reducing misdiagnosis rates significantly.
- **Faster Time-to-Treatment:** Expediting the diagnostic process, especially in remote areas.
- **Enhanced Clinical Decision Support:** Providing healthcare workers with a reliable assistant.
- **Data-Driven Insights:** Capturing valuable local health data for public health initiatives via SQLite logs.



Conclusion for your ML-Powered Medical Diagnosis Assistant project:

This project demonstrates the practical application of machine learning in enhancing healthcare accessibility, particularly in offline and resource-constrained environments. By leveraging interpretable ML models, statistical validation, and lightweight mobile integration, the system offers a scalable and ethical alternative to AI-heavy solutions. The use of offline speech recognition, Spark-based EHR matching, and confidence scoring ensures both usability and reliability without compromising patient privacy.